Homework 03

1.) $f(x) : \mathbb{R}^2 \to \mathbb{R}$, $\nabla f(\bar{x}) = [0 \quad 0]$

$$\nabla^2 f(\bar{x}) = \begin{bmatrix} 3 & -5 \\ -5 & 4 \end{bmatrix}$$ This is a saddle

a.) Find direction Δx that $f(\bar{x} + \varepsilon \Delta x) < f(\bar{x})$ holds

$$f(\bar{x} + \varepsilon \Delta x) = f(\bar{x}) + \underbrace{\cancel{\nabla f(\bar{x}) \varepsilon \Delta x}}_{0} + \frac{1}{2} \varepsilon^2 \Delta x^T \nabla^2 f(\bar{x}) \Delta x$$

$$= f(\bar{x}) + \frac{1}{2} \varepsilon^2 \Delta x^T \nabla^2 f(\bar{x}) \Delta x$$

For $f(\bar{x} + \varepsilon \Delta x) < f(\bar{x})$

$$\Rightarrow \Delta x^T \nabla^2 f(\bar{x}) \Delta x < 0$$

$\Delta x = [1 \quad 1]$

$$[1 \quad 1] \begin{bmatrix} 3 & -5 \\ -5 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = [-2 \quad 1] \begin{bmatrix} 1 \\ 1 \end{bmatrix} = -1$$

$-1 < 0$

b.) We need $\Delta x^T \nabla^2 f(\bar{x}) \Delta x > 0$

$\Delta x = [0, -1]$

$$[0 \quad -1] \begin{bmatrix} 3 & -5 \\ -5 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ -1 \end{bmatrix} = [5 \quad -4] \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1$$

2.) $f(x) : \mathbb{R} \to \mathbb{R}$ over $[-3, 4]$

$\varepsilon = 10^{-4}$

$L_k \leq 10^{-4} \Rightarrow L_0 \cdot \phi^k \leq 10^{-4}$

$\phi = 0.618$

$L_0 \Rightarrow \log L_0 \cdot \phi^k \leq \log(10^{-4})$

$\log L_0 + k \log \phi \leq -4 \log 10$

$k \log \phi \leq -4 \log 10 - \log L_0$

$k \leq \dfrac{-4 \log 10 - \log L_0}{\log \phi}$

$k \leq \dfrac{-4 \log 10 - \log 7}{\log \phi} \approx 23.18$

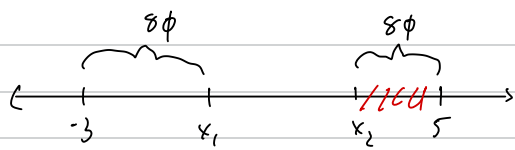
24 iterations

3.) $\max_{x \in \mathbb{R}} \quad x - 3e^x = f(x)$

$\varepsilon \leq 1$

s.t. $-3 \leq x \leq 5$

$r = 0.618$



$x_1 = a + (1-r)(b-a)$

$x_2 = a + r(b-a)$

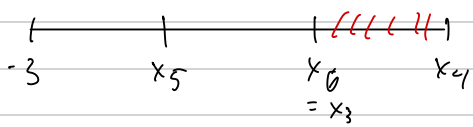
$8\phi = 4.944 \qquad x_1 = 0.056 \qquad x_2 = 1.944$

$f(x_1) = -3.116$

$f(x_2) = -10.016$

$x_3 = -1.111 \qquad f(x_3) = -2.09$

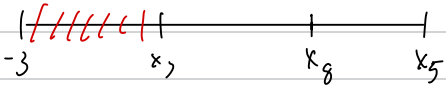
$x_4 = 0.056 \qquad f(x_4) = -3.116$



$x_5 = -1.83 \qquad f(x_5) = -2.313$

$x_6 = x_3 = -1.111 \qquad f(x_6) = -3.116$

$\varepsilon = 3.056$



$x_7 = -2.553$ $\qquad$ $f(x_7) = -2.786$

$x_8 = x_5 = -1.83$ $\qquad$ $f(x_8) = -2.313$

$\varepsilon = 0.723$ $\rightarrow$ Stop iteration

x^* is any point between $[-2.553, -1.83]$

$a^2+b^2 \geq 2ab$

$a^2+b^2 \geq \frac{(a+b)^2}{2}$

5.) a.) $f(x_1, x_2) = x_1^4 + x_2^4$

$\|x\|^2 = x_1^2 + x_2^2$

Since $x_1^{2^2} + x_2^{2^2} \geq \frac{x_1^2 + x_2^2}{2} = \frac{\|x\|^2}{2}$

Then $\lim_{\|x\| \to \infty} f(x_1, x_2) \to \infty$

$\therefore$ Coercive

b.) $f(x_1, x_2) = e^{x_1^2} + e^{x_2^2} - x_1^{40} - x_2^{30}$

Since exponential growth >>> polynomial growth
this is coercive

c.) $f(x_1, x_2) = 2x_1^2 - 8x_1x_2 + x_2^2$

$= x_1^2 + (x_1^2 + x_2^2) - 8x_1x_2$

$= \|x\|^2 + x_1^2 - \underbrace{8x_1x_2}$

$x_1^2 + x_2^2 \geq 2x_1x_2 \Rightarrow 4\|x\|^2 \geq 8x_1x_2$

$\leq \|x\|^2 - 4\|x\|^2 + x_1^2 = x_1^2 - 3\|x\|^2$

Since as $\|x\| \to \infty$, the $-3\|x\|^2$ terms goes to $-\infty$.
If we assume that this function is coercive, then
if $x_1 = 0$ and $x_2 \to \infty$, $f(x_1, x_2) \to -\infty$,
raising a contradiction

$\therefore$ not Coercive

d.) $f(x_1, x_2, x_3) = x_1^3 + x_2^3 + x_3^3$

$\|x\| = x_1^2 + x_2^2 + x_3^2$

If x_2 and $x_3 = 0$, and $x_1 \to -\infty$
then $\|x\| \to \infty$. However $f(x_1, x_2, x_3) \to -\infty$.

Therefore the function is not coercive